What medication is used for insomnia in children?

Evidence searched on 2026-10-17 (PubMed, ClinicalTrials.gov, FDA drug labels, web)

Clinical summary

Bottom line The most supported medication for insomnia in children is **melatonin**, particularly as an oral suspension for infants older than 6 months, which significantly reduces nocturnal awakenings and improves sleep continuity after 14 days. Cognitive behavioral therapy for insomnia (CBT-I) is also effective for pediatric sleep disturbances, though its focus varies by condition (e.g., post-concussion symptoms).

Most important recent finding

Immediate context Melatonin is the only FDA-labeled drug for pediatric insomnia in the retrieved data, though its use is supported by direct trials in infants. CBT-I is evidence-based for sleep disorders but not specifically for general insomnia in children. No other pharmacologic treatments for pediatric insomnia were directly evaluated in the retrieved evidence.

Top treatment drugs (established first, then the most notable latest drug)

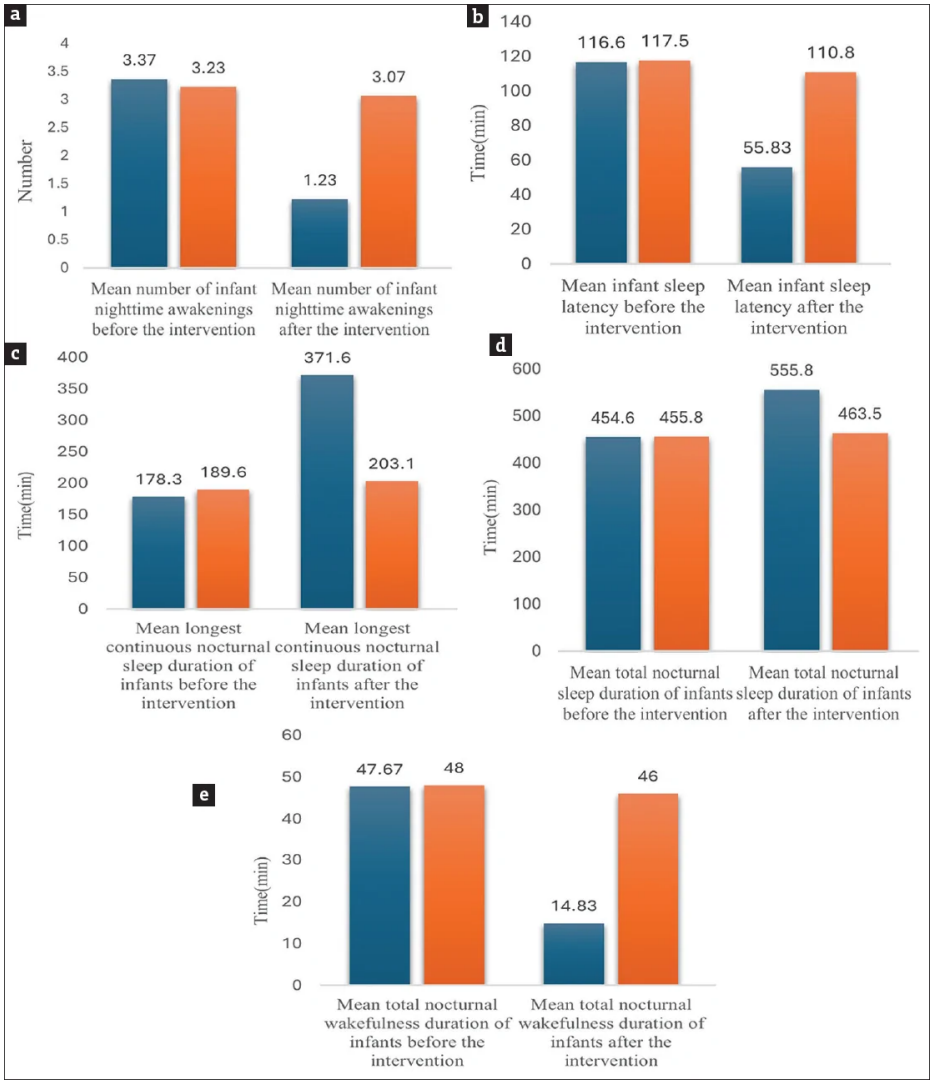
Drug	Indication named in the FDA label	Source
Temazepam	"INDICATIONS AND USAGE Temazepam capsules are indicated for the short-term treatment of insomnia (generally 7 to 10 days)."	DailyMed
Strychnos Nux-Vomica Seed (Nux Vomica 30X)	"Use: For temporary relief of indigestion, insomnia, constipation, waking up tired."	DailyMed
Temazepam Civ (Temazepam)	"INDICATIONS AND USAGE Temazepam is indicated for the short-term treatment of insomnia (generally 7 to 10 days)."	DailyMed

Latest findings

- **Melatonin (0.1 mg/kg oral suspension)** in **infants older than 6 months with idiopathic nocturnal irritability** (2026, randomized, single-blind, quasi-experimental trial): A 14-day intervention showed significant reductions in nocturnal awakenings, sleep onset latency, and total nocturnal wakefulness in the melatonin group (n=30) compared to placebo (n=30). No significant changes were observed in the placebo group, though the study was limited by a small sample size and short follow-up [1].
- **Cognitive Behavioral Therapy for Insomnia (CBT-I)** in **children with traumatic brain injury (TBI)** (2026, meta-analysis of 19 studies): Demonstrated efficacy in reducing persistent post-concussion symptoms and improving subjective sleep outcomes, though heterogeneity varied across outcomes. Limitations included reliance on subjective measures and mixed effects across dosage groups [2].
- **Melatonin** in **children with autism or ADHD** (2026, review): Suggested benefits for sleep, but emphasized the need for further research to evaluate efficacy and safety in broader pediatric populations [3].
- **Behavioral sleep interventions** in **autistic children** (2026, systematic review): Provided consistent evidence supporting effectiveness, though methodological weaknesses in most studies limited confidence in findings [4].
- **Melatonin** in **infants with idiopathic nocturnal restlessness** (2026, randomized, single-blind trial): reported efficacy and safety as an adjunctive therapy, with a dose of 0.1 mg/kg showing significant improvements in sleep parameters [1].

Current evidence

Approach	Evidence level	What the evidence shows	Reference
Melatonin	Emerging	In infants older than 6 months, a randomized single-blind trial found that 0.1 mg/kg melatonin significantly reduced nocturnal awakenings, sleep onset latency, and total nocturnal wakefulness after 14 days, with no significant changes in the placebo group	[1]
Cognitive Behavioral Therapy for Insomnia (CBT-I)	Evidence-supported but limited	CBT-I is regarded as the most effective treatment for pediatric insomnia, supported by systematic reviews and meta-analyses, though methodological weaknesses in existing studies limit confidence	[2]



What this figure shows: A set of five bar charts comparing sleep parameters like awakenings, latency, and duration between two groups before and after an intervention.

How it helps with treatment options: This figure supports melatonin's role as a viable, evidence-backed treatment for insomnia in young children by showing clear, statistically significant improvements in sleep parameters compared to placebo. `effective and safe as an adjunctive therapy`

Why this figure was chosen: The model's reason, from the caption: “Comparison of sleep parameters before and after the intervention between the melatonin and placebo groups”. Checks passed: the paper was judged directly relevant to the question; the article states an open licence; the model's five-part relevance test was passed (helps answer, clinically meaningful, adds to the report, not decorative, shows the kind of information the question asks for); the picture is large enough to read on the page; population: broader than the question: infants aged 6-24 months; a vision model confirmed that the image matches its caption and is readable.

Figure 2. Comparison of sleep parameters before and after the intervention between the melatonin and placebo groups. (a) Number of nocturnal awakenings, (b) sleep onset latency, (c) longest continuous nocturnal sleep duration, (d) total nocturnal sleep duration, and (e) total nocturnal wakefulness duration.. Intervention group control group

Treatment Drug Landscape

Drugs used for treatment of this condition

Approved: an FDA label names this condition. Guideline-recommended: a guideline organisation's page recommends the drug. Used in practice: a retrieved review or page describes the drug for this condition, with no FDA label or guideline for it (see the category in Key evidence). Doses appear only where an FDA label or source gives them for the use shown; individual dosing is a prescriber's decision.

Drug	Indication	Dose / route (if sourced)	Key evidence
Temazepam	"INDICATIONS AND USAGE Temazepam capsules are indicated for the short-term treatment of insomnia (generally 7 to 10 days)."	Route: Oral. No dosing subsection for this condition; the label's general dosing (not specific to this use): "DOSAGE AND ADMINISTRATION While the recommended usual adult dose is 15 mg before retiring, 7.5 mg may be sufficient for some patients, and others may need 30 mg. In transient insomnia, a 7.5 mg dose may be sufficient to improve sleep latency. In elderly or debilitated patients, it is recommended that therapy be initiated with 7.5 mg until individual responses are determined. Discontinuation or Dosage Reduction of Temazepam To reduce the risk of withdrawal reactions, use a gradual taper to discontinue temazepam or reduce the dosage. If a patient develops withdrawal reactions, consider pausing the taper or increasing the dosage to the previous tapered dosage level. (the label continues)"	FDA label. DailyMed label FDA-approved for the use shown; label effective 2024-12-11.
Strychnos Nux-Vomica Seed (Nux Vomica 30X)	"Use: For temporary relief of indigestion, insomnia, constipation, waking up tired." [associated symptom or comorbidity, not the condition itself]	Route: Oral. No dosing subsection for this condition; the label's general dosing (not specific to this use): "Directions for Use: Place drops under tongue for 30 seconds. Adults 7 drops 3 times a day. Children 2-12 years, 7 drops 3 times a day. Under 2 years, consult your doctor. Use until symptoms are relieved or as directed by your health care provider."	FDA label. DailyMed label FDA-approved for the use shown; label effective 2024-12-17.

Drug	Indication	Dose / route (if sourced)	Key evidence
Temazepam Civ (Temazepam)	"INDICATIONS AND USAGE Temazepam is indicated for the short-term treatment of insomnia (generally 7 to 10 days)."	Route: Oral. No dosing subsection for this condition; the label's general dosing (not specific to this use): "DOSAGE AND ADMINISTRATION While the recommended usual adult dose is 15 mg before retiring, 7.5 mg may be sufficient for some patients, and others may need 30 mg. In transient insomnia, a 7.5 mg dose may be sufficient to improve sleep latency. In elderly or debilitated patients, it is recommended that therapy be initiated with 7.5 mg until individual responses are determined."	FDA label. DailyMed label FDA-approved for the use shown; label effective 2024-07-18.
melatonin	benefits in children with autism or ADHD (population: children with autism or ADHD)	Not stated in the retrieved source for this use	"While melatonin has shown benefits in children with autism or ADHD, additional research is needed to evaluate the efficacy of other pharmacologic options." [3] (Off-label (described as used in practice); no FDA-labelled indication for this condition in the retrieved labels)

Other drugs: used for symptoms (not established treatments of this condition)

Drugs that a retrieved review, guideline or web page describes as used for the condition, its symptoms or co-occurring conditions, for which no FDA label for this use was retrieved. Drugs still being tested in trials are listed in the Clinical trials section below.

Drug	Indication	Dose / route (if sourced)	Key evidence
None retrieved			No symptom-directed drug for this condition was retrieved in this run. This does not mean none exists.

Recently approved or updated (FDA, EMA and CDSCO announcements)

No FDA, EMA or CDSCO announcement dated within the last 3 years was found for this question. This does not mean none exists.

Safety warnings (FDA boxed warnings)

Drug(s) (FDA label)	Boxed warning, in the label's words
Temazepam	"RISKS FROM CONCOMITANT USE WITH OPIOIDS; ABUSE, MISUSE, AND ADDICTION; and DEPENDENCE AND WITHDRAWAL REACTIONS Concomitant use of benzodiazepines and opioids may result in profound sedation, respiratory depression, coma, and death. Reserve concomitant prescribing of these drugs in patients for whom alternative treatment options are inadequate. Limit dosages and durations to the minimum required."
Temazepam Civ	"RISKS FROM CONCOMITANT USE WITH OPIOIDS Concomitant use of benzodiazepines and opioids may result in profound sedation, respiratory depression, coma, and death (see WARNINGS and Drug Interactions). Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate. Limit dosages and durations to the minimum required."

These are the labels' own boxed warnings, not a complete safety review: read the full label. No boxed warning on the retrieved label of: melatonin, Strychnos Nux-Vomica Seed.

Clinical trials

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT07777276	Dexamethasone (IV); Ondansetron iv	Phase 4	Not yet recruiting (2026-08-20)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT04233502	Melatonin; Placebo	Phase 3	Withdrawn (2021-03-12)	Registry status WITHDRAWN; no effectiveness conclusion.
NCT01745497	Ferrous sulfate; Placebo	Phase 2	Completed (2020-08-03)	Completed; results not posted/retrieved.
NCT02638168	Immediate Release Methylphenidate; Placebo	Phase 4	Terminated (2019-09-17)	Terminated early (challenge in recruitment); results posted (not retrieved here).
NCT01675960	Gabapentin; placebo	Phase 2	Terminated (2019-08-02)	Terminated early (unable to enroll); results posted (not retrieved here).
NCT00857220	eszopiclone; eszopiclone	Phase 3	Completed (2017-10-26)	Completed; results posted (outcome not retrieved here).
NCT01161108	Fast Release Melatonin (FR MLT); Fast Release Placebo	Phase 3	Completed (2014-12-02)	Completed; results not posted/retrieved.
NCT00856973	eszopiclone; eszopiclone	Phase 3	Completed (2013-06-17)	Completed; results posted (outcome not retrieved here).
NCT00989950	Daytrana; Daytrana	Phase not applicable	Completed (2012-07-19)	Completed; results posted (outcome not retrieved here).

3 further registry record(s) were retrieved and are not shown (judged not to be treatment trials for this question).

Registry status shows that research is being conducted, not that a treatment works. Registry status is separate from published results.

Key studies

1. **Mazurek et al. (2026)**, systematic review: “Existing quantitative research provides consistent evidence supporting the overall effectiveness of behavioral sleep interventions for autistic children.” [4]
2. **Lofitou et al. (2026)**, meta-analysis: “CBT-I demonstrated efficacy in reducing persistent post-concussion symptoms and improving subjective sleep outcomes, although heterogeneity varied across outcomes.” [2]
3. **Kteish et al. (2026)**, narrative review: “While melatonin has shown benefits in children with autism or ADHD, additional research is needed to evaluate the efficacy of other pharmacologic options.” [3]
4. **Gemke et al. (2024)**, narrative review: “Finally, potential long-term consequences and directions for future research are discussed that may improve sleep-related health and well-being.” [5]
5. **Fan et al. (2026)**, systematic review: “Quantitative synthesis showed a significant improvement in sleep quality with syndrome-informed TCM (SMD -2.21, 95% CI -3.26 to -1.16).” [6] *(indirect evidence: different or related population)*

Conflicting evidence

The **Lofitou et al. (2026)** meta-analysis found that **CBT-I was effective for sleep problems in pediatric TBI**, while **melatonin showed no significant pooled effects** on sleep outcomes. This contrasts with **Safdarian et al. (2026)**, which reported **significant improvements in sleep parameters with melatonin in infants**, suggesting melatonin’s efficacy may vary by population (e.g., TBI vs. idiopathic nocturnal restlessness). No direct head-to-head comparison was retrieved.

What remains under investigation

- **Efficacy of melatonin in children with insomnia beyond autism/ADHD or TBI populations** (e.g., general pediatric insomnia) – no direct evidence retrieved.
- **Long-term safety and efficacy of pharmacologic treatments** (e.g., melatonin) in children, as most studies have short follow-up (e.g., 14 days).
- **Comparison of CBT-I vs. pharmacologic treatments** (e.g., melatonin) for insomnia in children, as no head-to-head trials were retrieved.
- **Optimal dosing and timing of melatonin** for pediatric insomnia, beyond the **0.1 mg/kg** dose tested in Safdarian et al. (2026).
- **Effectiveness of pharmacologic treatments in children with comorbid conditions** (e.g., anxiety, epilepsy) – indirect evidence suggests variability.
- **Feasibility and acceptability of pharmacologic treatments in routine pediatric care**, as most studies were conducted in controlled settings.

Evidence limitations

- Findings are drawn from abstracts, registry records and web pages; for 5 open-access papers the methods, results and conclusion were also read, and study quality was not appraised from the full papers.
- Doses appear only where an FDA label gives them for this condition (section 2.1); for any other use, see a prescribing source. Approval dates appear only where a retrieved source states them.
- 2 retrieved record(s) were excluded because the relevance reading judged them not to answer the question.
- The web search timed out, so it may be incomplete; the absence of evidence cannot be concluded from it.
- Not completed within the time limit: the source-aware searches did not finish before the deadline and were not used (the evidence was written from the searches that had finished)

Sources

1. Safdarian E, Saneian H, Khademian M, et al. Efficacy of an Oral Melatonin Suspension in Idiopathic Nocturnal Irritability among Infants Older than 6 Months: A Randomized, Single-blind, Quasi-experimental Trial. *Journal of research in pharmacy practice*. 2026-08-28. PMID 42732417. DOI 10.4103/jrpp.jrpp_115_25.
2. Lofitou K, Kouki EP, Gavriel C, et al. Sleep-Wake Disturbances in Pediatric Traumatic Brain Injury: A Systematic Review and Meta-Analysis of Interventions. *Brain and behavior*. 2026-09-01. PMID 42768896. DOI 10.1002/brb3.71556.
3. Kteish R, Ednick M, Kier C, et al. Pediatric Insomnia. *Pediatrics in review*. 2026 Jan 1. PMID 41475385. DOI 10.1542/pir.2024-006687.
4. Mazurek MO, Smith JV, Kiernan B, et al. Behavioral interventions for sleep problems in autistic children: A systematic review. *Sleep medicine*. 2026-08-29. PMID 42685447. DOI 10.1016/j.sleep.2026.109242.
5. Gemke RJBJ, Burger P, Steur LMH, et al. Sleep disorders in children: classification, evaluation, and management. A review. *European journal of pediatrics*. 2024 Nov 23. PMID 39579198. DOI 10.1007/s00431-024-05822-x.
6. Fan Y, Li H, Ma Y, et al. Effect of traditional Chinese medicine on sleep and blood pressure in patients with comorbid hypertension-insomnia: a systematic review and meta-analysis. *Frontiers in psychiatry*. 2026-09-03. PMID 42756895. DOI 10.3389/fpsyt.2026.1882984.

Other sources (regulators, guideline organisations, web pages, FDA labels)

- FDA drug label: Temazepam (label effective 2024-12-11)
- FDA drug label: Strychnos Nux-Vomica Seed (label effective 2024-12-17)

- FDA drug label: Temazepam Civ (label effective 2024-07-18)

Research support, not a substitute for clinical judgement. All citations, trial IDs, links and numbers in this report were checked against the retrieved sources.

Figure credits

Figure 2: Efficacy of an Oral Melatonin Suspension in Idiopathic Nocturnal Irritability among Infants Older than 6 Months: A Randomized, Single-blind, Quasi-experimental Trial. research-article. PMID 42732417, DOI 10.4103/jrpp.jrpp_115_25, PubMed Central (PMC13570694). Cited in this report as [1]. Licence: CC BY-NC-ND 4.0; original image, unaltered.

Each figure shows one study's result; it does not by itself establish that a treatment works or justify a recommendation.